AI 101 for Mechanical Engineers · Module 8

Classes, scores and thresholds

Lesson 8.1

What you will learn

> Distinguish classes from scores, interpret the four confusion-matrix outcomes, and explain how a threshold affects fault alerts.

Lesson overview

Module 8 · Lesson 1

Classes, scores and thresholds

AI 101 for Mechanical Engineers

1 Turn a score into a decision

Model score
An output related to a class

→

Compare with threshold
The cutoff for a decision

→

Predict fault or healthy
A class label

Predict fault when score ≥ threshold.

A score between 0 and 1 is not automatically a calibrated probability. Check the score definition and calibration evidence.

2 Choose the threshold using validation

- Lowering the threshold makes more examples eligible for a fault alert.
- This usually catches more faults but can also raise more false alarms.
- Counts stay unchanged if no scores cross the new cutoff.
- Evaluate the chosen threshold on a separate final test.

3 Declare the positive class

Fault = positive. Healthy = negative.

These are evaluation labels, not judgements about whether an outcome is good or bad.

	Predicted fault	Predicted healthy
Actual fault	**TP: true positive** Detected fault	**FN: false negative** Missed fault
Actual healthy	**FP: false positive** False alarm	**TN: true negative** Correct healthy prediction

The matrix records counts for a stated threshold and dataset. Read the actual and predicted headings before interpreting a cell; their positions can differ between diagrams.

Key takeaway — **Distinguish scores, probabilities and decisions. Use validation evidence to choose a cutoff.**

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A threshold turns a score into a decision



- A binary detector distinguishes two classes
- Label fault when score ≥ threshold

A score is not automatically a probability

- Some outputs are probabilities; others are decision values [40]
- A value between zero and one is not automatically calibrated
- Check the score’s definition and calibration evidence first
- Choosing a cutoff and showing calibration are distinct questions

> Here calibration means stated probabilities agree with observed frequencies, unlike sensor calibration (Module 3).

Fault is positive; healthy is negative

Outcome	Actual	Predicted
True positive (TP)	Fault	Fault
False negative (FN)	Fault	Healthy: missed fault
False positive (FP)	Healthy	Fault: false alarm
True negative (TN)	Healthy	Healthy

These are evaluation labels, not judgements about whether an outcome is desirable.

Read confusion-matrix headings explicitly

- Counts are for a stated threshold and evaluation dataset [41]
- Actual and predicted heading positions can differ between diagrams
- TP, FN, FP and TN keep the same meanings

Lowering the threshold changes the decisions

Lower threshold

- More examples eligible for a fault label
- Generally catches more faults

Possible cost

- Can produce more false alarms
- Need not change every count

Decisions change even when the scores stay fixed.

Select on validation, then test

1. Weigh missed-fault and false-alarm consequences
2. Choose the threshold on validation examples
3. Evaluate the selected threshold on the final test
4. Do not keep tuning it using test labels

CHECK YOUR UNDERSTANDING

If fault is positive, what does a false negative represent?

Answer before moving on.

CHECK YOUR UNDERSTANDING

If fault is positive, what does a false negative represent?

> An actual fault predicted as healthy: a missed fault.

CHECK YOUR UNDERSTANDING

Does a score of 0.9 alone establish a fault probability?

Answer before moving on.

CHECK YOUR UNDERSTANDING

Does a score of 0.9 alone establish a fault probability?

> No. Its definition and calibration evidence are still needed before treating it as a probability.

Key takeaways

- Distinguish a score, a probability and a thresholded decision
- Declare the positive class before reading counts
- FN means a missed fault; FP means a false alarm
- Select thresholds using validation evidence and error consequences

References

[40] A. Niculescu-Mizil and R. Caruana, “Predicting good probabilities with supervised learning,” in Proc. 22nd Int. Conf. Mach. Learn. (ICML), 2005, pp. 625–632. doi: 10.1145/1102351.1102430

[41] T. Fawcett, “An introduction to ROC analysis,” Pattern Recognit. Lett., vol. 27, no. 8, pp. 861–874, 2006. doi: 10.1016/j.patrec.2005.10.010

Numbers match the course reference list. The written lesson is the authoritative text.